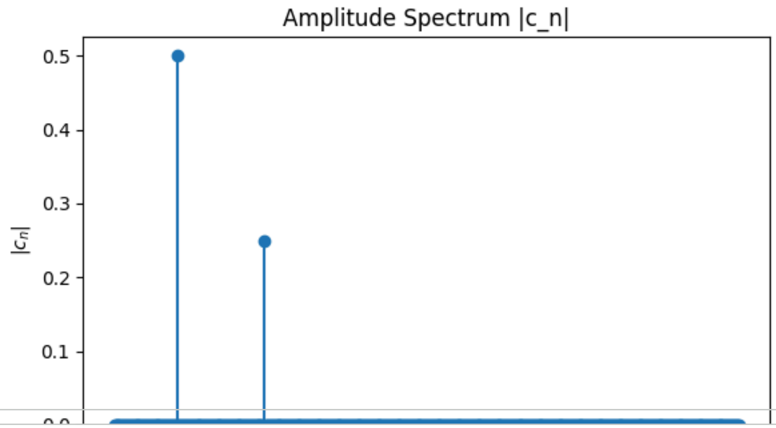
```
import numpy as np


T = 10
n = 1000
t = np.linspace(0, T, n+1)
f = np.sin(t)

h = T / n

y = f**2

simpson_integral = (h/3) * (
    y[0] +
    y[-1] +
    4 * np.sum(y[1:-1:2]) +
    2 * np.sum(y[2:-1:2])
)

mean_square_amplitude = simpson_integral / T

print(f"Mean-square amplitude = {mean_square_amplitude:.6f}")
```

```
Mean-square amplitude = 0.477176
```

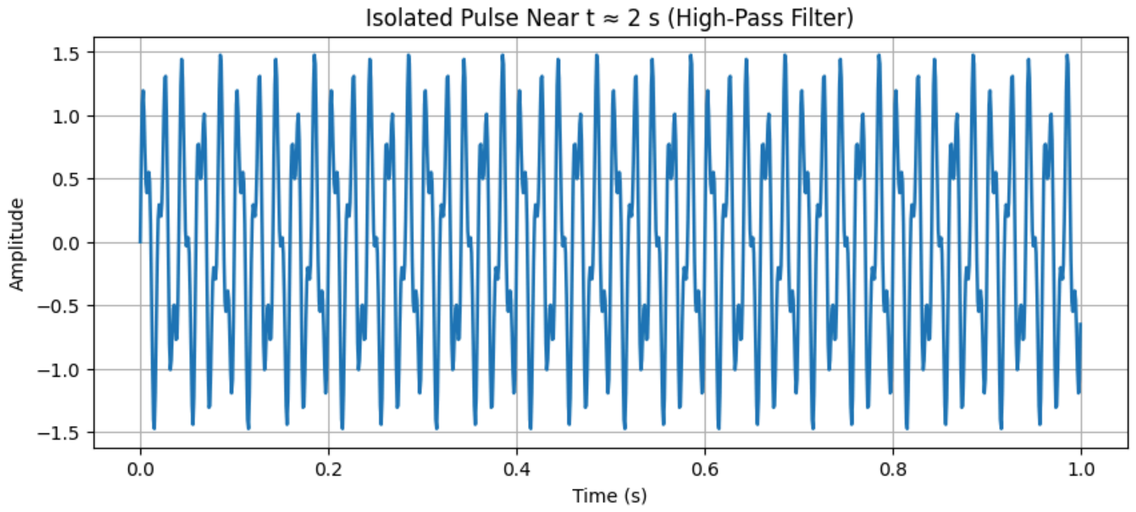
```
import matplotlib.pyplot as plt
import numpy as np

T = 1.0
N = 1000
t = np.linspace(0, T, N, endpoint=False)
f = np.sin(2*np.pi*50*t) + 0.5*np.sin(2*np.pi*120*t)

dt = t[1] - t[0]
c = np.fft.fft(f) / N
freq = np.fft.fftfreq(N, d=dt)

Cmag = np.abs(c)
pos = freq >= 0

plt.figure(figsize=(6,4))
plt.stem(freq[pos], Cmag[pos], basefmt=" ")
plt.xlabel('Frequency (Hz)')
plt.ylabel(r'$|c_n|$')
plt.title('Amplitude Spectrum |c_n|')
plt.tight_layout()
plt.show()
```




```
energy_spectrum = np.sum(np.abs(c)**2)
mean_sq = mean_square_amplitude
print(f"Parseval check: {mean_sq:.6f} vs. {energy_spectrum:.6f}")
```

```
Parseval check: 0.477176 vs. 0.625000
```

```
import numpy as np
import matplotlib.pyplot as plt


N = len(f)
dt = t[1] - t[0]
freq = np.fft.fftfreq(N, dt)

c = np.fft.fft(f)

fc = 1.5
H = (np.abs(freq) > fc).astype(float)

c_hp = c * H


f_hp = np.fft.ifft(c_hp).real

plt.figure(figsize=(10,4))
plt.plot(t, f_hp, linewidth=1.8)
plt.title("Isolated Pulse Near t ≈ 2 s (High-Pass Filter)")
plt.xlabel("Time (s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
```




```
import numpy as np
from scipy.signal import stft, istft

T = 5.0
N = 5000
t = np.linspace(0, T, N, endpoint=False)

f = np.sin(2*np.pi*1.5*t) + 0.3*np.sin(2*np.pi*10*t)
f += np.exp(-200*(t-2)**2) * np.sin(2*np.pi*30*t)

dt = t[1] - t[0]
fs = 1 / dt

f_stft, t_stft, Z = stft(f, fs=fs, nperseg=int(0.15 / dt))

k = np.argmin(np.abs(t_stft - 2.0))

mask = np.zeros_like(Z)

i1 = max(k - 1, 0)
i2 = min(k + 2, Z.shape[1])

freq_mask = (f_stft > 3)[:, None]

mask[:, i1:i2] = freq_mask

Z_pulse = Z * mask

_, pulse_est = istft(Z_pulse, fs=fs)
pulse_est = pulse_est[:len(f)]

f_clean = f - pulse_est
```

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